Title: **A congruence obstruction to Heuristic 1 in POKÉ-2D**

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This note concerns the probability estimates in *On weak keys of POKÉ* [1]. For POKÉ-2D, the square-free part of $D'^2 - \delta$ satisfies congruence restrictions absent from the uniform reference model in Heuristic 1. We give an explicit statistical-distance bound and an admissible candidate set with zero actual weak-key probability but a constant-order estimate in the paper's regime $D'^2 \asymp \delta_m$. For the actual KeyGen parameters, the square condition targeted by Algorithm 1 is impossible. Detailed proofs are in the appendices.

1 Setting and the distributional obstruction

We use the paper's notation:

$$\Delta = \{q(A-q) : q \in \{1, \dots, A-1\}\}, \qquad \delta_m = \max \Delta = \frac{A^2}{4}.$$

Assume $8 \mid A$, which holds in the original POKÉ-2D parameterization [2] with $A = 2^a$ and $a \geq 3$. We retain the paper's conditions $\gcd(A, D) = 1$, $D' \mid D$, and $D' \geq \sqrt{\delta_m}$. Thus D' is odd and $D'^2 > \delta_m$.

Let $\mathrm{sf}(n)$ denote the unique square-free integer r such that $n = rc^2$ for a positive integer c. Let

$$\delta \xleftarrow{\$} \Delta, \qquad \delta_0 \xleftarrow{\$} \{1, \dots, \delta_m\}, \qquad S = \mathrm{sf}(D'^2 - \delta), \quad S_0 = \mathrm{sf}(D'^2 - \delta_0).$$

Write $\mathbb{P}_\Delta$ and $\mathbb{P}_\mathrm{U}$ for these two probability models. The first sampling rule is uniform over the distinct degrees in Δ, as in the paper. For square-free r, Heuristic 1 asserts the unquantified approximation

$$\mathbb{P}_\Delta[S = r] \approx \mathbb{P}_\mathrm{U}[S_0 = r]. \tag{1}$$

Here $\mathcal{L}(S)$ denotes the probability distribution of S. The total variation distance is

$$\begin{aligned} \mathrm{TV}(\mathcal{L}(S), \mathcal{L}(S_0)) &:= \sup_{\mathcal{E} \subseteq \mathbb{N}} |\mathbb{P}_\Delta[S \in \mathcal{E}] - \mathbb{P}_\mathrm{U}[S_0 \in \mathcal{E}]| \\ &= \frac{1}{2} \sum_{r \geq 1} |\mathbb{P}_\Delta[S = r] - \mathbb{P}_\mathrm{U}[S_0 = r]|. \end{aligned}$$

Proposition 1 (Explicit distributional discrepancy)**.** *Let $\varepsilon = (\lfloor \log_4 \delta_m \rfloor + 2)/\delta_m$. The degree set Δ imposes the restriction*

$$S \bmod 8 \in \{1, 2, 5\}.$$

In contrast, for each $u \in \{1, 2, 3, 5, 6, 7\}$,

$$\left| \mathbb{P}_\mathrm{U}[S_0 \equiv u \pmod 8] - \frac{1}{6} \right| \leq \varepsilon.$$

Consequently, for the event $\mathcal{B} = \{r \in \mathbb{N} : r \bmod 8 \in \{3, 6, 7\}\}$,

$$\mathbb{P}_\Delta[S \in \mathcal{B}] = 0, \qquad \left| \mathbb{P}_\mathrm{U}[S_0 \in \mathcal{B}] - \frac{1}{2} \right| \leq 3\varepsilon, \tag{2}$$

and hence

$$\mathrm{TV}(\mathcal{L}(S), \mathcal{L}(S_0)) \geq \frac{1}{2} - 3\varepsilon. \tag{3}$$

Since $\varepsilon \to 0$ as $\delta_m \to \infty$, Heuristic 1 cannot hold in total variation with vanishing error. If interpreted only as an additive $o(1)$ error for each fixed r, it is not contradicted by this argument. However, that interpretation neither controls the relative error in the small probabilities estimated in Theorem 14 nor justifies summing the errors over a growing set $\mathcal{R}$ in Theorem 17.

2 An admissible candidate set for Theorem 17

Definition 16 requires only that $\mathcal{R} \subseteq \{1, \ldots, \delta_m\}$ consist of square-free integers. In particular, it permits

$$\mathcal{R} = \{1 \le r \le \delta_m : r \text{ is square-free}, \ r \equiv 3 \pmod 4\}. \tag{4}$$

This is a parameter-dependent family of sets. The restriction to the smallest square-free integers is introduced separately in Section 4.2 of the paper and is not a hypothesis of Theorem 17.

Denote the probability expression displayed in Theorem 17 by

$$F(\mathcal{R}) = \frac{1}{D' + \sqrt{D'^2 - \delta_m}} \sum_{r \in \mathcal{R}} \frac{1}{\sqrt{r}}. \tag{5}$$

Proposition 2 (An allowed set with zero actual probability)**.** *For the set* (4)*, under any degree distribution supported on* Δ*,*

$$\mathbb{P}_\Delta[\exists r \in \mathcal{R}, \ c \in \mathbb{N} : \ D'^2 - \delta = rc^2] = 0. \tag{6}$$

Nevertheless,

$$F(\mathcal{R}) = \frac{4}{\pi^2} \frac{\sqrt{\delta_m}}{D' + \sqrt{D'^2 - \delta_m}} + O\left(\frac{\log \delta_m}{D' + \sqrt{D'^2 - \delta_m}}\right). \tag{7}$$

Thus, in the regime $D'^2 \asymp \delta_m$*,* $F(\mathcal{R}) = \Theta(1)$*. In particular, if* $D'^2/\delta_m \to 1$*, then* $F(\mathcal{R}) \to 4/\pi^2 \approx 0.4053$*.*

In that regime, the exact uniform reference probability for the same set is also bounded away from zero; see Appendix C. Thus the failure of the reference model is not merely an artefact of approximating integer counts.

The expression in Theorem 17 therefore cannot approximate the actual POKÉ-2D probability with additive error $o(1)$ uniformly over all its allowed candidate sets. This identifies a limitation in applying the heuristic estimate; it does not refute a conditional implication whose premise is Heuristic 1. Moreover, $\#\mathcal{R} = \Theta(\delta_m)$, so this example does not invalidate the conditional recovery algorithm or establish an attack contradicting the proposed security parameters. The actual probability being smaller also leaves a one-sided upper bound unrefuted.

3 Effect of the KeyGen coprimality conditions

Remark. In the original POKÉ-2D parameterization [2], valid secret degrees additionally satisfy $\gcd(\delta, AD) = 1$. Since A is even, this forces q to be odd. Hence

$$\delta = q(A - q) \equiv 7 \pmod 8, \qquad D'^2 - \delta \equiv 2 \pmod 8, \qquad \mathrm{sf}(D'^2 - \delta) \equiv 2 \pmod 8.$$

For any sampling weights on these admissible degrees, the event $\mathrm{sf}(D'^2 - \delta) \not\equiv 2 \pmod 8$ has probability zero. Keeping the paper's unconditioned reference distribution unchanged, its probability differs from $5/6$ by at most ε. Thus the statistical distance lower bound strengthens to

$$\frac{5}{6} - \varepsilon.$$

The zero-probability statement for (4) remains valid.

In particular, Algorithm 1 of [1], based on Theorem 7, targets the condition

$$D'^2 - \deg \varphi = c^2.$$

This would require $\mathrm{sf}(D'^2 - \deg \varphi) = 1$, contradicting the congruence above. Hence no valid POKÉ-2D secret key satisfies this condition for any admissible $D' \mid D$: the square-condition

weak-key class targeted by Algorithm 1 has probability zero. This does not exclude the generalized case in Algorithm 2 with $r \equiv 2 \pmod 8$.

A revised probability model should account for these local restrictions. The approximation in Heuristic 1 also needs an explicit error criterion and a range of candidate sets over which errors may be summed. Accounting for these congruence restrictions is necessary, but does not by itself justify the probability estimates.

A Proof of the distributional discrepancy

For $\delta = q(A-q)$, odd D' and $8 \mid A$ give

$$D'^2 - \delta \equiv 1 + q^2 \pmod 8 \in \{1, 2, 5\}.$$

In particular, $4 \nmid D'^2 - \delta$. Writing $D'^2 - \delta = rc^2$ with r square-free, c must be odd; therefore $c^2 \equiv 1 \pmod 8$ and $r \equiv D'^2 - \delta \pmod 8$.

For the reference model, every positive integer n has a unique decomposition $n = 4^j m$ with $4 \nmid m$. Since the square factor remaining in m is odd, $\mathrm{sf}(n) \equiv m \pmod 8$. Consequently, for $u \in \{1, 2, 3, 5, 6, 7\}$,

$$\mathrm{sf}(n) \equiv u \pmod 8 \quad \Longleftrightarrow \quad n = 4^j(8k + u), \qquad j, k \geq 0. \tag{8}$$

For fixed u, the sets indexed by j are disjoint.

Now $D'^2 - \delta_0$ is uniform on the δ_m integers in

$$I = \{D'^2 - \delta_m, \ldots, D'^2 - 1\}.$$

Set $M = D'^2 - 1$ and $J = \lfloor \log_4 \delta_m \rfloor$. For $0 \leq j \leq J$, let $h = 8 \cdot 4^j$ and $a = 4^j u$. The exact count is

$$\begin{aligned} \#\{n \in I : n \equiv a \pmod h\} &= \left\lfloor \frac{M-a}{h} \right\rfloor - \left\lfloor \frac{M - \delta_m - a}{h} \right\rfloor \\ &= \frac{\delta_m}{h} + e_{j,u}, \qquad |e_{j,u}| < 1. \end{aligned} \tag{9}$$

Indeed, $e_{j,u}$ is the difference of the fractional parts of $(M - \delta_m - a)/h$ and $(M-a)/h$. No divisibility assumption is needed. All remaining terms, with $j > J$, are divisible by 4^{J+1}. Since $4^{J+1} > \delta_m$, at most one integer in I can contribute to this tail. Denote its contribution for residue u by $T_u \in \{0, 1\}$. It follows that

$$\mathbb{P}_{\mathrm{U}}[S_0 \equiv u \pmod 8] = \frac{1}{6}(1 - 4^{-(J+1)}) + \frac{\sum_{j=0}^{J} e_{j,u} + T_u}{\delta_m}.$$

Now $0 < \delta_m/(6 \cdot 4^{J+1}) < 1/6$, so

$$\left| T_u - \frac{\delta_m}{6 \cdot 4^{J+1}} \right| < 1.$$

Combining this with $|e_{j,u}| < 1$ gives

$$\left| \mathbb{P}_{\mathrm{U}}[S_0 \equiv u \pmod 8] - \frac{1}{6} \right| \leq \frac{J+2}{\delta_m} = \varepsilon.$$

This bound is independent of the position of I and requires no upper bound on D'^2/δ_m. Summing the classes $3, 6, 7$ gives (2). Finally, total variation is the supremum of the probability difference over events, giving (3).

B Square-free counting and the weighted sum

Lemma 1. *For $x \geq 2$, let*

$$N(x) = \#\{n \leq x : n \text{ is square-free}, \ n \equiv 3 \pmod 4\}.$$

Then

$$N(x) = \frac{2}{\pi^2}x + O(\sqrt{x}), \qquad \sum_{\substack{n \leq x \\ n \text{ square-free} \\ n \equiv 3 \ (4)}} \frac{1}{\sqrt{n}} = \frac{4}{\pi^2}\sqrt{x} + O(\log x). \tag{10}$$

Proof. Let μ be the Möbius function. The identity

$$\mu(n)^2 = \sum_{a^2 \mid n} \mu(a)$$

follows by expanding $\prod_{p^2 \mid n}(1-1)$: the empty product is 1, whereas the product is 0 if a squared prime divides n. Interchanging finite sums gives

$$N(x) = \sum_{a \leq \sqrt{x}} \mu(a) \#\{b \leq x/a^2 : a^2 b \equiv 3 \pmod 4\}.$$

Even a contribute nothing. For odd a, $a^2 \equiv 1 \pmod 4$, and

$$\#\{b \leq y : b \equiv 3 \pmod 4\} = \left\lfloor \frac{y+1}{4} \right\rfloor = \frac{y}{4} + O(1).$$

Consequently,

$$N(x) = \frac{x}{4} \sum_{\substack{a \leq \sqrt{x} \\ a \text{ odd}}} \frac{\mu(a)}{a^2} + O(\sqrt{x}). \tag{11}$$

The error follows from $|\mu(a)| \leq 1$ and at most $\sqrt{x}$ summands. Also,

$$\sum_{a > \sqrt{x}} \frac{|\mu(a)|}{a^2} = O(x^{-1/2}),$$

so completing the sum in (11) to infinity changes the answer by $O(\sqrt{x})$.

For completeness, the relevant Euler product follows from

$$\prod_{\substack{3 \leq p \leq y \\ p \text{ prime}}} (1 - p^{-2}) = \sum_{\substack{a \geq 1, \ a \text{ odd} \\ p \mid a \Rightarrow p \leq y}} \frac{\mu(a)}{a^2}.$$

Expanding the finite product selects distinct odd primes, and their product a has coefficient $\mu(a)$. The omitted terms have $a > y$, with absolute sum at most $\sum_{a>y} a^{-2} \to 0$. Hence

$$\sum_{\substack{a \geq 1 \\ a \text{ odd}}} \frac{\mu(a)}{a^2} = \prod_{p \neq 2} (1 - p^{-2}) = \frac{\zeta(2)^{-1}}{1 - 2^{-2}} = \frac{8}{\pi^2}.$$

Substituting into (11) proves the first formula.

To prove the weighted formula, let $a_n = \mathbf{1}_{\{n \text{ square-free}, n \equiv 3 \ (4)\}}$. Then $N(t) = \sum_{n \leq t} a_n$. For $f \in C^1([1, x])$, the identity $f(n) = f(x) - \int_n^x f'(t)\,dt$ and linearity of integration over a finite sum imply

$$\begin{aligned} \sum_{n \leq x} a_n f(n) &= N(x)f(x) - \int_1^x \left(\sum_{n \leq x} a_n \mathbf{1}_{\{n \leq t\}} \right) f'(t)\,dt \\ &= N(x)f(x) - \int_1^x N(t) f'(t)\,dt. \end{aligned}$$

Applying this exact identity to $f(t) = t^{-1/2}$ yields

$$\sum_{n \le x} \frac{a_n}{\sqrt{n}} = \frac{N(x)}{\sqrt{x}} + \frac{1}{2} \int_1^x N(t) t^{-3/2}\, dt.$$

Write $N(t) = \alpha t + E(t)$, where $\alpha = 2/\pi^2$ and $E(t) = O(\sqrt{t})$ for $t \ge 1$. The endpoint contributes $\alpha\sqrt{x} + O(1)$, and the main term in the integral contributes

$$\frac{\alpha}{2} \int_1^x t^{-1/2}\, dt = \alpha(\sqrt{x} - 1).$$

The remaining integral is bounded in absolute value by

$$\frac{1}{2} \int_1^x |E(t)| t^{-3/2}\, dt = O\left(\int_1^x \frac{dt}{t} \right) = O(\log x).$$

Combining these terms proves the second formula. □

C The candidate-set counterexample and reference counts

For $\delta \in \Delta$,

$$D'^2 - \delta \equiv 1 + q^2 \pmod 4 \in \{1, 2\}.$$

For $r \equiv 3 \pmod 4$, however, rc^2 is 0 or 3 modulo 4. Thus the equality $D'^2 - \delta = rc^2$ is impossible, proving (6) for any sampling weights on Δ.

By Lemma 1,

$$\#\mathcal{R} = \frac{2}{\pi^2}\delta_m + O(\sqrt{\delta_m}), \qquad \sum_{r \in \mathcal{R}} \frac{1}{\sqrt{r}} = \frac{4}{\pi^2}\sqrt{\delta_m} + O(\log \delta_m).$$

Substitution in (5) and division of numerator and denominator by $\sqrt{\delta_m}$ prove (7). If $D'^2 \asymp \delta_m$, the denominator in (7) is bounded above and bounded below by 1, whereas the numerator tends to $4/\pi^2$. This proves the constant-order assertion. If $D'^2/\delta_m \to 1$, the denominator tends to 1.

Exact reference probability. Let $b = \lceil D'/\sqrt{\delta_m} \rceil$ and count only reference degrees with $D'^2 - \delta_0 = b^2 r$, $r \in \mathcal{R}$. Such r lie in

$$\left[\frac{D'^2 - \delta_m}{b^2}, \frac{D'^2 - 1}{b^2} \right] \subset (0, \delta_m].$$

Every square-free $r \equiv 3 \pmod 4$ in this interval yields a distinct valid δ_0 and satisfies $\mathrm{sf}(b^2 r) = r$. Lemma 1, with an $O(1)$ endpoint correction, gives

$$\#\{r \text{ in this interval} : r \text{ square-free}, \ r \equiv 3 \pmod 4\} = \frac{2(\delta_m - 1)}{\pi^2 b^2} + O(\sqrt{\delta_m}).$$

For endpoints below 2, the same estimate follows by absorbing a bounded error. Dividing by δ_m proves

$$\mathbb{P}_{\mathrm{U}}[S_0 \in \mathcal{R}] \ge \frac{2}{\pi^2 b^2} - O(\delta_m^{-1/2}). \tag{12}$$

Thus the exact reference distribution already assigns positive constant mass, when $D'^2 \asymp \delta_m$ and hence $b = O(1)$, to a set that is impossible under the actual degree support.

Rounding and accumulation of errors. For each square-free r, the exact reference probability is

$$\begin{aligned}\mathbb{P}_{\mathrm{U}}[S_0 = r] &= \frac{1}{\delta_m}\left(\left\lceil \frac{D'}{\sqrt{r}} \right\rceil - \left\lceil \sqrt{\frac{D'^2 - \delta_m}{r}} \right\rceil\right) \\ &= \frac{1}{\sqrt{r}(D' + \sqrt{D'^2 - \delta_m})} + \eta_r, \qquad |\eta_r| \le \frac{1}{\delta_m}.\end{aligned}$$

Define the model error $\varepsilon_r = \mathbb{P}_{\Delta}[S = r] - \mathbb{P}_{\mathrm{U}}[S_0 = r]$. For any admissible $\mathcal{R}$,

$$\mathbb{P}_{\Delta}[S \in \mathcal{R}] = F(\mathcal{R}) + \sum_{r\in\mathcal{R}} \varepsilon_r + \sum_{r\in\mathcal{R}} \eta_r, \qquad \left|\sum_{r\in\mathcal{R}} \eta_r\right| \le \frac{\#\mathcal{R}}{\delta_m}. \tag{13}$$

Pointwise additive $o(1)$ bounds on ε_r do not bound their sum over growing sets. For the family (4), the available rounding bound is also constant-sized. Statements (2) and (12) establish the model discrepancy independently of these rounding errors.

References

[1] Tomoki Moriya. *On weak keys of POKÉ*. Cryptology ePrint Archive, Paper 2026/1002, version dated 2026-05-21. https://eprint.iacr.org/2026/1002.

[2] Andrea Basso and Luciano Maino. *POKÉ: A Compact and Efficient PKE from Higher-dimensional Isogenies*. EUROCRYPT 2025, pp. 94–123. https://eprint.iacr.org/2024/624.